N1 — Apex Unitaries: An Explicit Family in U(N−1) over Multiquadratic–Cyclotomic Compositums

Leonardo Murillo Montero

May 28, 2026

Abstract

For each integer $N \geq 2$ we study the unitary matrix $U_N \in U(N-1)$ that intertwines the normalized Helmert basis and the nontrivial additive characters of $\mathbb{Z}/N$ on the mean-zero hyperplane of $\mathbb{C}^N$. Its entries lie in the compositum $K_N \cdot \mathbb{Q}(\zeta_N)$ of the multiquadratic field $K_N = \mathbb{Q}(\sqrt{p} : p \leq N)$ and the cyclotomic field $\mathbb{Q}(\zeta_N)$, and split as a product of a quadratic-radical row scalar and a cyclotomic integer. We describe this arithmetic structure, compute the degree of the compositum, and determine the basic spectral invariants at $N = 4$ in closed form: the determinant is i, the trace is an explicit element of $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$ whose field trace over $\mathbb{Q}(\zeta_4)$ vanishes, and the characteristic polynomial is given explicitly; in particular not all eigenvalues of U_4 are roots of unity, although their product is. Finally we describe the action of the Galois group of the compositum on U_N as a signed monomial transformation $U_N \mapsto D\,U_N\,P$, and use it to explain the rationality of $\det U_4$ over $\mathbb{Q}(\zeta_4)$. The determinant for general N is obtained in closed form in the companion paper.

Contents

1. Introduction . . . 2
2. The apex unitary . . . 3
   2.1 Two orthonormal frames of the hyperplane . . . 3
   2.2 Definition and entry formula . . . 3
   2.3 The arithmetic field and the radical–cyclotomic split . . . 3
   2.4 The degree of the compositum . . . 4
3. Invariants at level four . . . 4
   3.1 The determinant . . . 4
   3.2 The trace and its field trace . . . 5
   3.3 The characteristic polynomial and the spectrum . . . 5
4. The Galois action on U_N . . . 6
   4.1 The Galois group of the compositum . . . 6
   4.2 The cyclotomic action: column permutations . . . 6
   4.3 The multiquadratic action: row sign changes . . . 6
   4.4 Cyclotomic rationality of the determinant . . . 6
5. Further directions . . . 7

References . . . 7

1. Introduction

Fix $N \geq 2$ and work on the mean-zero hyperplane

$$H_N := \left\{ v \in \mathbb{C}^N : \sum_{j=0}^{N-1} v_j = 0 \right\}$$

with the standard Hermitian inner product $\langle u, v \rangle = \sum_j u_j \overline{v_j}$. The space H_N carries two natural orthonormal frames: the real *Helmert frame* of nested contrasts, and the complex *character frame* of nontrivial additive characters of $\mathbb{Z}/N$. The unitary change of basis between them is a matrix $U_N \in U(N-1)$ whose entries are explicit algebraic numbers in the compositum $K_N \cdot \mathbb{Q}(\zeta_N)$, combining quadratic-radical and cyclotomic content.

This paper sets up the family $\{U_N\}_{N \geq 2}$ as an arithmetic object and records its first invariants. Section 2 gives the definition, the field of definition, the factorization of the entries into a radical part and a cyclotomic part, and the degree of the compositum. Section 3 computes the determinant, trace, and characteristic polynomial at $N = 4$ in closed form; a noteworthy point is that the trace fails to be an algebraic integer, which forces at least one eigenvalue of U_4 to be a non-root-of-unity even though the product of the eigenvalues is the root of unity i. Section 4 describes the action of $\mathrm{Gal}(K_N \cdot \mathbb{Q}(\zeta_N)/\mathbb{Q})$ on U_N as a signed monomial transformation and uses it to explain why $\det U_4$ lies in the cyclotomic field $\mathbb{Q}(\zeta_4)$ despite the radicals carried by the individual entries.

The determinant for general N is determined in closed form in the companion paper [N2]: $\det U_N = i^{-\binom{N-1}{2}}$. Section 5 lists the invariants taken up in subsequent papers of the series.

Throughout, $\omega = e^{2\pi i/N}$ and $\zeta_m = e^{2\pi i/m}$; $\pi(N)$ is the number of primes $\leq N$ and φ is Euler's totient.

---

2. The apex unitary

2.1 Two orthonormal frames of the hyperplane

The **Helmert vectors** h_k ($k = 1, \dots, N-1$) consist of k leading ones, then the entry $-k$, then $N-1-k$ trailing zeros:

$$h_k := (\,1, \dots, 1,\, -k,\, 0, \dots, 0\,) \in \mathbb{Z}^N.$$

They are real, mutually orthogonal, mean-zero, and satisfy $\|h_k\|^2 = k(k+1)$; the normalized vectors $e_k := h_k/\sqrt{k(k+1)}$ form an orthonormal basis of H_N.

The **character vectors** are $\chi_c := (\omega^{cj})_{j=0}^{N-1}$ for $c = 1, \dots, N-1$. They are mean-zero and orthogonal with $\|\chi_c\|^2 = N$, and the normalized $\tilde{\chi}_c := \chi_c/\sqrt{N}$ form a second orthonormal basis of H_N.

2.2 Definition and entry formula

Definition 2.1. The *apex unitary at level* N is the change-of-basis matrix $U_N \in \mathbb{C}^{(N-1)\times(N-1)}$ with entries

$$(U_N)_{kc} := \langle \tilde{\chi}_c, e_k \rangle = \frac{1}{\sqrt{Nk(k+1)}} \Big(\sum_{j=0}^{k-1} \omega^{cj} - k\,\omega^{ck} \Big), \qquad k, c \in \{1, \dots, N-1\}.$$

Because e_k is real, $(U_N)_{kc} = R_k \sum_j h_k(j)\,\omega^{cj}$ with $R_k := 1/\sqrt{Nk(k+1)}$. Summing the geometric series gives the equivalent closed form (T9)

$$(U_N)_{kc} = R_k \cdot \frac{1 - (k+1)\,\omega^{kc} + k\,\omega^{(k+1)c}}{1 - \omega^c}. \qquad (2.1)$$

Since U_N is the matrix of an orthonormal change of basis, it is unitary: $U_N \in U(N-1)$.

Remark 2.2 (orientation of the pairing). The determinant is sensitive to the conjugation convention. Definition 2.1 pairs the character $\tilde{\chi}_c$ against the real Helmert vector e_k, i.e. it is conjugate-linear in the character; equivalently it pairs h_k bilinearly with $\chi_c(j) = \omega^{cj}$. This is the convention reproduced by the closed form (2.1) and by the explicit matrix (3.1) below. The opposite pairing $\langle e_k, \tilde{\chi}_c \rangle = \overline{(U_N)_{kc}}$ yields the complex-conjugate matrix $\overline{U_N}$, also unitary, with conjugate determinant. We fix the convention of Definition 2.1 once and for all.

2.3 The arithmetic field and the radical–cyclotomic split

Let $K_N := \mathbb{Q}(\sqrt{p} : p \le N \text{ prime})$ be the multiquadratic field generated by the square roots of the primes up to N, and let $\mathbb{Q}(\zeta_N)$ be the N-th cyclotomic field. The compositum $L_N := K_N \cdot \mathbb{Q}(\zeta_N)$ contains every entry of U_N.

Proposition 2.3 (radical–cyclotomic split). Each entry factors as

$$(U_N)_{kc} = R_k\, C_{kc}, \qquad R_k = \frac{1}{\sqrt{Nk(k+1)}} \in K_N, \qquad C_{kc} = \frac{1 - (k+1)\omega^{kc} + k\omega^{(k+1)c}}{1 - \omega^c} \in \mathbb{Z}[\omega] \subset \mathbb{Q}(\zeta_N). \qquad (2.2)$$

The radical factor R_k depends only on the row index k and lies in the real multiquadratic field; the cyclotomic factor C_{kc} is an algebraic integer in $\mathbb{Q}(\zeta_N)$.

Proof. Immediate from (2.1): R_k is a real quadratic radical, and clearing the denominator $1 - \omega^c$ in the geometric sum shows $C_{kc} = \sum_{j=0}^{k-1} \omega^{cj} - k\,\omega^{ck} \in \mathbb{Z}[\omega]$. □

2.4 The degree of the compositum

The two factors of L_N overlap in a multiquadratic field: a square root $\sqrt{m}$ (for $m \mid \prod_{p \le N} p$ squarefree) lies in $\mathbb{Q}(\zeta_N)$ exactly when the conductor of $\mathbb{Q}(\sqrt{m})$ divides N, where $\operatorname{cond}(\mathbb{Q}(\sqrt{m})) = m$ if $m \equiv 1 \pmod 4$ and $4m$ otherwise. The set of such m forms an $\mathbb{F}_2$-vector space under multiplication modulo squares; let $t(N)$ be its dimension, so that $[K_N \cap \mathbb{Q}(\zeta_N) : \mathbb{Q}] = 2^{t(N)}$. By the degree formula for composita,

$$d(N) := [L_N : \mathbb{Q}] = \frac{[K_N : \mathbb{Q}]\,[\mathbb{Q}(\zeta_N) : \mathbb{Q}]}{[K_N \cap \mathbb{Q}(\zeta_N) : \mathbb{Q}]} = \frac{2^{\pi(N)}\,\varphi(N)}{2^{t(N)}}.$$

Table 2.1. The compositum degree for small N.

N	$\pi(N)$	$\varphi(N)$	$t(N)$	$K_N \cap \mathbb{Q}(\zeta_N)$	$d(N)$
2	1	1	0	$\mathbb{Q}$	2
3	2	2	0	$\mathbb{Q}$	8
4	2	2	0	$\mathbb{Q}$	8
5	3	4	1	$\mathbb{Q}(\sqrt{5})$	16
6	3	2	0	$\mathbb{Q}$	16
7	4	6	0	$\mathbb{Q}$	96
8	4	4	1	$\mathbb{Q}(\sqrt{2})$	32
30	10	8	1	$\mathbb{Q}(\sqrt{5})$	4096

The sequence $d(N)$ is irregular: $d(7) = 96$ dwarfs its neighbours because the real subfield of $\mathbb{Q}(\zeta_7)$ is cubic and therefore shares no quadratic subfield with any multiquadratic K_N, so $t(7) = 0$.

---

3. Invariants at level four

We compute the determinant, trace, and characteristic polynomial of U_4. Here $\omega = i$ and, writing out (2.1),

$$U_4 = \begin{pmatrix} \dfrac{1-i}{2\sqrt{2}} & \dfrac{1}{\sqrt{2}} & \dfrac{1+i}{2\sqrt{2}} \\[2mm] \dfrac{3+i}{2\sqrt{6}} & \dfrac{-1}{\sqrt{6}} & \dfrac{3-i}{2\sqrt{6}} \\[2mm] \dfrac{i}{\sqrt{3}} & \dfrac{1}{\sqrt{3}} & \dfrac{-i}{\sqrt{3}} \end{pmatrix}. \tag{3.1}$$

Its entries lie in $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i) = L_4$, of degree 8 over $\mathbb{Q}$ (here $\sqrt{6} = \sqrt{2}\,\sqrt{3}$).

3.1 The determinant

Proposition 3.1. $\det U_4 = i$.

Proof. Factor the common radical out of each row of (3.1): pulling $1/(2\sqrt{2})$, $1/(2\sqrt{6})$, $1/\sqrt{3}$ from rows $1, 2, 3$ respectively leaves the integer matrix

$$M = \begin{pmatrix} 1-i & 2 & 1+i \\ 3+i & -2 & 3-i \\ i & 1 & -i \end{pmatrix},$$

and the product of the extracted scalars is $\frac{1}{2\sqrt{2}} \cdot \frac{1}{2\sqrt{6}} \cdot \frac{1}{\sqrt{3}} = \frac{1}{4\sqrt{36}} = \frac{1}{24}$. Expanding $\det M$ along the third row,

$$\det M = i \begin{vmatrix} 2 & 1+i \\ -2 & 3-i \end{vmatrix} - \begin{vmatrix} 1-i & 1+i \\ 3+i & 3-i \end{vmatrix} - i \begin{vmatrix} 1-i & 2 \\ 3+i & -2 \end{vmatrix} = i(8) - (-8i) - i(-8) = 24i.$$

Hence $\det U_4 = \frac{1}{24} \det M = \frac{24i}{24} = i$. $\square$

The radicals $\sqrt{2}, \sqrt{3}, \sqrt{6}$ that appear throughout (3.1) cancel completely: $\det U_4 = i \in \mathbb{Q}(\zeta_4)$ carries no multiquadratic content. Section 4.4 gives the structural reason; the general statement, $\det U_N = i^{-\binom{N-1}{2}}$, is proved in [N2], of which Proposition 3.1 is the case $N = 4$.

3.2 The trace and its field trace

Summing the diagonal of (3.1) and rationalizing,

$$\operatorname{tr} U_4 = \frac{1-i}{2\sqrt{2}} - \frac{1}{\sqrt{6}} - \frac{i}{\sqrt{3}} = \frac{(3\sqrt{2} - 2\sqrt{6}) - i\,(3\sqrt{2} + 4\sqrt{3})}{12}. \tag{3.2}$$

Numerically $\operatorname{tr} U_4 \approx -0.0547 - 0.9310\,i$, consistent with $|\operatorname{tr} U_4| < \sqrt{3}$.

Thus $\operatorname{tr} U_4$ has genuine multiquadratic content: it lies in $L_4 \setminus \mathbb{Q}(\zeta_4)$. Nevertheless its field trace over the cyclotomic part vanishes.

Proposition 3.2. $\operatorname{Tr}_{L_4/\mathbb{Q}(\zeta_4)}(\operatorname{tr} U_4) = 0$.

Proof. The extension $L_4/\mathbb{Q}(\zeta_4)$ has degree 4, since $K_4 = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ is degree 4 over $\mathbb{Q}$ and $K_4 \cap \mathbb{Q}(\zeta_4) = \mathbb{Q}$ (the field $\mathbb{Q}(\zeta_4) = \mathbb{Q}(i)$ is imaginary quadratic and contains no real quadratic field). Its Galois group $\operatorname{Gal}(L_4/\mathbb{Q}(\zeta_4)) \cong (\mathbb{Z}/2)^2$ fixes i and acts by independent sign changes $\sqrt{2} \mapsto \pm\sqrt{2}$, $\sqrt{3} \mapsto \pm\sqrt{3}$. Summing the four conjugates of (3.2), each of $\sqrt{2}, \sqrt{3}, \sqrt{6}$ occurs with both signs and cancels; hence the field trace is 0. $\square$

3.3 The characteristic polynomial and the spectrum

For a 3×3 matrix, $\chi_{U_4}(t) = t^3 - (\operatorname{tr} U_4)\, t^2 + (\operatorname{tr} \bigwedge^2 U_4)\, t - \det U_4$. For a unitary matrix $\operatorname{tr} \bigwedge^2 U_4 = \det U_4 \cdot \operatorname{tr}(U_4^{-1}) = \det U_4 \cdot \overline{\operatorname{tr} U_4}$. With $\det U_4 = i$ and (3.2),

$$\chi_{U_4}(t) = t^3 - \frac{(3\sqrt{2} - 2\sqrt{6}) - i\,(3\sqrt{2} + 4\sqrt{3})}{12}\, t^2 + \frac{-(3\sqrt{2} + 4\sqrt{3}) + i\,(3\sqrt{2} - 2\sqrt{6})}{12}\, t - i, \tag{3.3}$$

an explicit cubic with coefficients in L_4.

Proposition 3.3. Not every eigenvalue of U_4 is a root of unity, although the product of the eigenvalues is the root of unity $\det U_4 = i$.

Proof. A sum of roots of unity is an algebraic integer. The element $\operatorname{tr} U_4$ of (3.2) is not an algebraic integer: for instance the component $\sqrt{2}/4$ has minimal polynomial $16t^2 - 2$ over $\mathbb{Z}$, with leading coefficient > 1, so $\operatorname{tr} U_4 \notin \mathcal{O}_{L_4}$. If all three eigenvalues were roots of unity their sum $\operatorname{tr} U_4$ would be an algebraic integer; hence at least one eigenvalue is not a root of unity. The product of the eigenvalues equals $\det U_4 = i$ by Proposition 3.1. $\square$

The explicit factorization of (3.3), and with it the precise position of the three eigenvalues on the unit circle, is not pursued here.

---

4. The Galois action on U_N

The Galois group of the compositum acts on U_N entry by entry. Through the split (2.2) this action takes a transparent form: the cyclotomic part permutes columns, and the multiquadratic part rescales rows by signs.

4.1 The Galois group of the compositum

Write $L_N = K_N \cdot \mathbb{Q}(\zeta_N)$ and $G_N = \mathrm{Gal}(L_N/\mathbb{Q})$. Restriction gives a short exact sequence

$$1 \longrightarrow \mathrm{Gal}(L_N/\mathbb{Q}(\zeta_N)) \longrightarrow G_N \longrightarrow (\mathbb{Z}/N)^\times \longrightarrow 1,$$

where $\mathrm{Gal}(L_N/\mathbb{Q}(\zeta_N)) \cong \mathrm{Gal}(K_N/K_N \cap \mathbb{Q}(\zeta_N)) \cong (\mathbb{Z}/2)^{\pi(N)-t(N)}$ consists of sign changes on the quadratic radicals not already in $\mathbb{Q}(\zeta_N)$, and $(\mathbb{Z}/N)^\times = \mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ acts by $\omega \mapsto \omega^a$. At $N = 4$ one has $G_4 \cong (\mathbb{Z}/2)^2 \times (\mathbb{Z}/4)^\times \cong (\mathbb{Z}/2)^3$ of order $8 = d(4)$, generated by $\sigma_2 : \sqrt{2} \mapsto -\sqrt{2}$, $\sigma_3 : \sqrt{3} \mapsto -\sqrt{3}$, and $\sigma_{-1} : i \mapsto -i$.

4.2 The cyclotomic action: column permutations

Let $\sigma \in G_N$ restrict to $\omega \mapsto \omega^a$, $a \in (\mathbb{Z}/N)^\times$, and fix the radicals R_k (such σ exist precisely as the lifts that act trivially on K_N). Then $\sigma(R_k) = R_k$ and, from (2.2), $\sigma(C_{kc}) = \sum_j h_k(j)\,\omega^{acj} = C_{k,ac}$. Multiplication by the unit a permutes the index set $\{1, \dots, N-1\}$, so

$$\sigma(U_N) = U_N\, P_a,$$

where P_a is the permutation matrix of $c \mapsto ac$ on $\{1, \dots, N-1\} \subset \mathbb{Z}/N$.

At $N = 4$, $(\mathbb{Z}/4)^\times = \{1, 3\}$ and σ_{-1} ($a = 3$) swaps columns 1 and 3 of (3.1) while fixing column 2 (the real character $\chi_2(j) = (-1)^j$); direct inspection confirms $\sigma_{-1}(U_4) = U_4\, P_3$.

4.3 The multiquadratic action: row sign changes

Let $\tau \in \mathrm{Gal}(L_N/\mathbb{Q}(\zeta_N))$ fix ω and act by sign changes on the radicals. Then τ fixes each C_{kc} and sends $R_k \mapsto \varepsilon_k(\tau)\, R_k$ with $\varepsilon_k(\tau) \in \{\pm 1\}$, where $\varepsilon_k(\tau) = -1$ exactly when τ flips the squarefree part of $Nk(k+1)$. Hence

$$\tau(U_N) = D_\tau\, U_N, \qquad D_\tau = \mathrm{diag}\,(\varepsilon_1(\tau), \dots, \varepsilon_{N-1}(\tau)).$$

At $N = 4$ the row radicals are $R_1 = 1/(2\sqrt{2})$, $R_2 = 1/(2\sqrt{6})$, $R_3 = 1/(4\sqrt{3})$, so σ_2 gives $D = \mathrm{diag}(-1, -1, +1)$ and σ_3 gives $D = \mathrm{diag}(+1, -1, -1)$. For instance $\sigma_2((U_4)_{11}) = \sigma_2(\frac{1-i}{2\sqrt{2}}) = -\frac{1-i}{2\sqrt{2}}$, consistent with the sign on row 1.

4.4 Cyclotomic rationality of the determinant

Combining the two actions, a general $\sigma \in G_N$ acts by a signed monomial transformation

$$\sigma(U_N) = D_\sigma\, U_N\, P_\sigma,$$

with D_σ a diagonal sign matrix and P_σ a permutation matrix. Taking determinants,

$$\sigma(\det U_N) = \det(\sigma(U_N)) = \det(D_\sigma)\, \det(U_N)\, \mathrm{sgn}(P_\sigma).$$

In particular, $\det U_N$ is fixed by the multiquadratic subgroup $\mathrm{Gal}(L_N/\mathbb{Q}(\zeta_N))$—equivalently $\det U_N \in \mathbb{Q}(\zeta_N)$—precisely when $\det(D_\tau) = +1$ for every such τ.

At $N = 4$ this holds: $\det D = (-1)(-1)(+1) = +1$ for σ_2 and $\det D = (+1)(-1)(-1) = +1$ for σ_3. Thus $\det U_4$ is fixed by $\langle \sigma_2, \sigma_3 \rangle$ and lies in $\mathbb{Q}(\zeta_4)$, which is the structural reason behind the cancellation observed after Proposition 3.1. For general N the sign product $\det(D_\tau)$ need not be $+1$: it is shown in [N2] that $\det U_N \in \mathbb{Q}(\zeta_N)$ if and only if $N \not\equiv 3 \pmod 4$, the obstruction in the remaining class being a single non-cyclotomic quadratic radical.

---

5. Further directions

The family $\{U_N\}$ supports several arithmetic invariants beyond those computed here, each the subject of a subsequent paper in the series.

- **The determinant.** Determined in closed form in [N2]: $\det U_N = i^{-\binom{N-1}{2}}$, a fourth root of unity, lying in $\mathbb{Q}(\zeta_N)$ exactly when $N \not\equiv 3 \pmod 4$.
- **The trace.** The vanishing of Proposition 3.2 does *not* persist for all N. The general field trace is

  $$\mathrm{Tr}_{L_N/\mathbb{Q}(\zeta_N)}(\mathrm{tr}\, U_N) = 2^s \sum_{k \in S_N} (U_N)_{kk}, \qquad s = \pi(N) - t(N), \quad S_N = \{k : \sqrt{\mathrm{sf}(Nk(k+1))} \in \mathbb{Q}(\zeta_N)\},$$

  obtained by Galois sign-character orthogonality: each $\tau \in \mathrm{Gal}(L_N/\mathbb{Q}(\zeta_N))$ fixes C_{kk} and rescales R_k by a sign, so $\mathrm{Tr}_{L_N/\mathbb{Q}(\zeta_N)}\left((U_N)_{kk}\right) = 2^s (U_N)_{kk}$ for $k \in S_N$ and 0 otherwise. Thus the genuinely non-cyclotomic part $\sum_{k \notin S_N} (U_N)_{kk}$ *always* has vanishing field trace — the correct generalization of Proposition 3.2 — while the field trace of $\mathrm{tr}\, U_N$ itself vanishes iff $\sum_{k \in S_N} (U_N)_{kk} = 0$. The case $N = 4$ vanishes because $S_4 = \emptyset$; the first $N \geq 3$ with nonvanishing field trace is $N = 5$, where $\mathrm{Tr}_{L_5/\mathbb{Q}(\zeta_5)}(\mathrm{tr}\, U_5) = -2\zeta_5$. The power traces $\mathrm{tr}\, U_N^m$ governing the characteristic-polynomial coefficients are a natural next object.
- **The spectrum.** Proposition 3.3 shows the spectrum need not consist of roots of unity. Its conductor-block structure for composite N, and its description at prime N via Gauss sums, are natural next objects.
- **The Galois module.** The signed monomial action of Section 4 makes $\{U_N\}$ a module over G_N; its orbit structure and Galois-stable invariants connect the spectral data to the cyclotomic units and real class numbers of $\mathbb{Q}(\zeta_N)$.

References

[N2] Murillo Montero, L. (2026). *The Apex Determinant: A Closed Form via Helmert–Fourier Factorization.* First draft, June 5, 2026.

[T9] Murillo Montero, L. (2026). *The Helmert–Fourier Transfer Matrix.* First draft, May 26, 2026.

Lang, S. (2002). *Algebra.* Revised third edition. Graduate Texts in Mathematics 211. Springer.

Washington, L. C. (1997). *Introduction to Cyclotomic Fields.* Second edition. Graduate Texts in Mathematics 83. Springer.